RAJALAKSHMI ENGINEERING COLLEGE

Jeeva T 251801115-2025-AIDS

J2

Started on	Tuesday, 29 September 2026, 9:47 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 9:49 AM
Time taken	2 mins 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFirstZero(int arr[], int low, int high) {
    // No zero found
    if (low > high)
        return -1;

    // If low itself is the first zero
    if (arr[low] == 0) {
        if (low == 0 || arr[low - 1] == 1)
            return low;
    }

    int mid = (low + high) / 2;

    // Search left half if mid is 0
    if (arr[mid] == 0) {
        return findFirstZero(arr, low, mid - 1);
    }

    // If mid is 1, first zero must be in right half
    return findFirstZero(arr, mid + 1, high);
}

int main() {
    int m;
    scanf("%d", &m);

    int arr[m];

    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    int firstZero = findFirstZero(arr, 0, m - 1);

    if (firstZero == -1)
        printf("0\n");
    else
        printf("%d\n", m - firstZero);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on	Tuesday, 29 September 2026, 9:48 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 9:49 AM
Time taken	1 min 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int nums[], int n) {
    int candidate = nums[0];
    int count = 1;

    for (int i = 1; i < n; i++) {
        if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }

        if (count == 0) {
            candidate = nums[i];
            count = 1;
        }
    }

    return candidate;
}

int main() {
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    printf("%d\n", majorityElement(nums, n));
```

```
35      return 0;
36  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Jeeva T 251801115-2025-AIDS

J2

Started on	Tuesday, 29 September 2026, 9:48 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 9:49 AM
Time taken	28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x) {
    // No element <= x
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    // If arr[mid] is equal to x, it is the floor
    if (arr[mid] == x)
        return arr[mid];

    // If arr[mid] is greater than x,
    // search in the left half
    if (arr[mid] > x)
        return findFloor(arr, low, mid - 1, x);

    // arr[mid] < x
    // It can be the floor, but there may be a
    // larger valid element in the right half
    int result = findFloor(arr, mid + 1, high, x);

    if (result == -1)
        return arr[mid];

    return result;
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int floor = findFloor(arr, 0, n - 1, x);

    printf("%d\n", floor);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Started on	Tuesday, 29 September 2026, 9:50 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 9:50 AM
Time taken	29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int target) {
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == target)
        return mid;

    if (arr[mid] > target)
        return binarySearch(arr, low, mid - 1, target);

    return binarySearch(arr, mid + 1, high, target);
}

int findPair(int arr[], int n, int x, int i) {
    // All elements have been checked
    if (i >= n - 1)
        return 0;

    int target = x - arr[i];

    // Search for the required complement
    // only after index i
    int pos = binarySearch(arr, i + 1, n - 1, target);

    if (pos != -1) {
        printf("%d\n", arr[i]);
        printf("%d\n", arr[pos]);
        return 1;
    }

    // Check the next element
    return findPair(arr, n, x, i + 1);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    if (!findPair(arr, n, x, 0)) {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Jeeva T 251801115-2025-AIDS

Started on	Tuesday, 29 September 2026, 9:50 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 9:51 AM
Time taken	27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }

    swap(&arr[i + 1], &arr[high]);

    return i + 1;
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);

        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

int main() {
    int n;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);

        if (i < n - 1)
            printf(" ");
    }
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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